

AI-Powered Insurance Claims Assistant — API Documentation

Base URL: <http://127.0.0.1:8000/api/v1> | Interactive OpenAPI Docs: <http://127.0.0.1:8000/docs>

1. Health Check Endpoint

GET /health — Returns system readiness, vector index count, and model initialization status.

Sample Response:

```
{"status": "healthy", "service": "AI-Powered Insurance Claims Assistant", "vector_store_indexed_count": 500, "ml_models_ready": true}
```

2. Input Validation Guardrails

POST /claims/validate — Validates claim schema, range constraints, and sanitizes prompt injections.

Sample Request:

```
{"claim_type": "Auto", "state": "CA", "policyholder_tenure_years": 4.5, "previous_claims_count": 0, "incident_date": "2024-01-10", "claim_filed_date": "2024-01-14", "claim_amount": 3500.0, "deductible": 500.0}
```

Sample Response:

```
{"is_valid": true, "errors": [], "warnings": [], "sanitized_data": {...}}
```

3. Hybrid RAG Search Endpoint

POST /claims/hybrid-search — Retrieves similar historical claims using ChromaDB vector search + BM25 keyword matching with RRF scoring.

Sample Request:

```
{"query": "rear fender collision repair", "top_k": 3, "claim_type": "Auto", "state": "CA"}
```

Sample Response:

```
{"query": "rear fender collision repair", "count": 3, "results": [{"claim_id": "CLM-0000000001", "hybrid_score": 0.9412, "retrieval_method": "Hybrid", "claim_amount": 3200.0, "claim_status": "Approved"}]}
```

4. Classification & Prioritization Endpoint

POST /claims/classify — Classifies claim complexity and assigns SLA priority triage score.

Sample Response:

```
{"complexity_level": "Low / Standard Complexity", "priority_tier": "P4 - Low / Routine", "priority_score": 25.0, "target_sla_resolution_hours": 120}
```

5. Predictive Machine Learning Endpoint

POST /claims/predict — Predicts claim payout amount, resolution days, and fraud probability with explainability.

Sample Response:

```
{"predicted_claim_amount": 3680.0, "estimated_amount_range": "$3,128.00 - $4,416.00", "predicted_days_to_resolution": 14.2, "fraud_probability_percent": 18.5, "risk_tier": "Low Risk", "top_risk_drivers": ["Claim attributes match normal underwriting patterns."]}
```

6. Multi-Agent End-to-End Analysis Workflow

POST /claims/analyze — Executes the complete 3-agent orchestration (Investigation → Risk → Recommendation) and LLM-as-Judge audit.

Sample Response:

```
{"classification": {...}, "ml_predictions": {...}, "investigation": {"agent_name": "Investigation Agent", "findings": "..."}, "risk_assessment": {"fraud_probability_percent": 18.5}, "recommendation": {"decision": "AUTO_APPROVE", "fast_track_eligible": true, "recommended_payout": "$3,000.00"}, "llm_as_judge_review": {"verdict": "PASS", "overall_quality_score": 9.3}}
```

7. Adjuster Feedback Loop Endpoints

POST /claims/feedback — Records human adjuster reviews (Approve/Modify/SIU) into persistent store.

GET /claims/feedback/metrics — Returns total reviews and AI agreement rate analytics.